

User Instructions

Additional Features

Instruction	Description	14-bit Format
ADD Ra, Rb	Add values in Ra and Rb and store result in Ra ($Ra \leftarrow Ra + Rb$)	0000 RaRaRa RbRbRb 0000
NEG R	2's complement of R ($R \leftarrow -R$)	0001 RRR 000 0000
MOVI R, d	Move immediate value d to R ($R \leftarrow d$)	0010 RRR 000 dddd
JZR R, d	Jump if R == 0; else $PC \leftarrow PC + 1$	0011 RRR 000 dddd
COM Ra, Rb	Compare values in Ra and Rb	0100 RaRaRa RbRbRb 0000
MUL Ra, Rb	Multiply last two bits of Ra and Rb and store in Ra	0101 RaRaRa RbRbRb 0000
BWA Ra, Rb	Bitwise AND ($Ra \leftarrow Ra \& Rb$)	0110 RaRaRa RbRbRb 0000
BWO Ra, Rb	Bitwise OR ($Ra \leftarrow Ra Rb$)	0111 RaRaRa RbRbRb 0000
BWX Ra, Rb	Bitwise XOR ($Ra \leftarrow Ra \wedge Rb$)	1000 RaRaRa RbRbRb 0000
NOT R	Bitwise NOT of R ($R \leftarrow \sim R$)	1001 RRR 000 0000

OUTPUT DISPLAY

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LED OUTPUTS

The 4 LEDs display the binary value stored in Register 7 (R7).

Output[0] → Least Significant Bit

Output[1] → Bit 1

Output[2] → Bit 2

Output[3] → Most Significant Bit

SEVEN SEGMENT DISPLAY

The seven-segment display shows the hexadecimal value stored in Register 7 (R7).

Example Values:

R7 = 0000 → 0

R7 = 0011 → 3

R7 = 0110 → 6

R7 = 1111 → F

RESET OPERATION

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When reset is pressed:

- Program Counter resets to 0
- All registers are cleared
- Execution restarts from beginning
- Outputs are cleared